Evaluating Terrain Representation and Accuracy Across Multiple DEM Datasets: A Case Study at Timpanogos Cave National Monument

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Abstract

This study evaluates six (6) digital elevation models (DEMs) of varying spatial resolution and source origin to assess their effectiveness for terrain representation and modeling accuracy within Timpanogos Cave National Monument (TICA). The selected DEMs include lidar2m, dem1m, ned10m, ned30m, srtm30m, and srtm90m. Each dataset was analyzed for its ability to represent topographic features through derived slope and curvature surfaces, characterize local variability, and accurately predict elevation compared to 75 high-accuracy GPS points. High-resolution DEMs, particularly lidar2m and dem1m, aligned closely with known terrain features and produced the lowest absolute elevation errors. Coarser datasets, such as srtm90m and ned30m, generalized surface features and introduced substantial vertical error. Local variation analysis revealed that curvature is the most sensitive derivative to resolution changes, while elevation and slope retained more stability across scales. A clear relationship was also observed between terrain steepness and vertical error in several datasets. These findings highlight that no single DEM is optimal for all applications. Instead, the choice of elevation data should be guided by the intended analytical purpose, terrain complexity, and acceptable error thresholds. High-resolution DEMs are ideal for detailed, localized studies, while coarser products may suffice for broader-scale assessments where precision is less critical.

Introduction

Digital Elevation Models (DEMs) serve as foundational datasets in terrain analysis, supporting a wide array of environmental GIS modeling applications. These include hydrological modeling, habitat assessment, and other terrain-based applications. DEMs differ substantially in spatial resolution, data collection methodology, and accuracy. In turn, influencing the different types of analytical tasks that can be conducted with certain types of DEM datasets. These variations can affect the reliability and interpretation of derived terrain attributes such as slope and curvature (Wilson, 2012). Understanding the strengths and limitations of different DEMs is therefore critical to selecting the appropriate dataset for a study's operational scale and analytical goals (Cao & Lam, 1997). To explore how these variations manifest in practice, this study compares multiple DEMs of varying resolution and origin within a real-world landscape.

This study investigates six (6) DEMs representing the Timpanogos Cave National Monument (TICA), a topographically complex area in Utah's Wasatch Mountains (McNeil et al., 2002). The DEMs utilized in the study include: a custom 1-meter DEM created from digitized contour lines using the Topo to Raster tool; a 2-meter LiDAR DEM provided by Utah's AGRC; USGS National Elevation Dataset (NED) DEMs at 10-meter and 30-meter resolutions; and Shuttle Radar Topography Mission (SRTM) DEMs at 30-meter and 90-meter resolutions. Through a series of terrain analyses, such as slope and curvature derivations, local variability assessments, and GPS-based accuracy evaluations. This study compares the performance of each DEM with respect to terrain and absolute error.

By examining the interaction between resolution, source method, and topographic complexity. This study aims to evaluate what is gained and lost with each elevation dataset. Specifically, how each DEM performs in terms of local surface variation, reliability of terrain

derivatives, and the magnitude and distribution of vertical error. Insights from prior research, including Holmes et al. (2000), Woodcock and Strahler (1987), and McNeil et al. (2002), were utilized to contextualize the observed differences and explore implications for environmental modeling.

Material and Methods

1-Meter Topo to Raster DEM

A custom 1-meter resolution DEM was generated using the "Topo to Raster" interpolation tool within ArcGIS Pro (Esri, n.d.-a). This dataset was created from digitized vector contour lines representing elevation within the TICA study area. The Topo to Raster method is hydrologically enforced and designed to produce smooth and continuous elevation surfaces, making it suitable for terrain modeling in areas with limited original data (Wilson, 2012). The dem1m DEM was generated using a Projected Coordinate System (PCS) of NAD 1927 UTM Zone 12N and was clipped to the extent of the TICA GPS dataset. This DEM served as a high-resolution, locally derived product that reflects the original data's topography while introducing interpolation-based smoothing artifacts.

2-Meter LiDAR DEM

The 2-meter LiDAR DEM was sourced from the Utah Automated Geographic Reference Center (AGRC, 2008). It was derived by airborne LiDAR and was acquired between October 2006 and May 2007 by Horizons, Inc.. With sub-meter vertical accuracy and minimal interpolation, LiDAR DEMs are considered among the most accurate elevation products available, particularly in complex terrain (Woodcock & Strahler, 1987). The lidar2m DEM was

reprojected to a PCS of NAD 1927 UTM Zone 12N and was clipped to the extent of the TICA GPS dataset. This DEM was used as a high-quality reference surface for comparing elevation errors and topographic derivatives across scales.

10-Meter and 30-Meter NED DEMs

The 10-meter and 30-meter resolution DEMs were obtained from the USGS National Elevation Dataset (NED). A national gridded product compiled from multiple source inputs and resampled for consistency. The 10-meter NED is derived from higher-resolution source data and resampled using a bilinear method, while the 30-meter version reflects broader national coverage and coarser base inputs (U.S. Geological Survey, 2009c, 2009d). The ned10m and ned30m DEMs were reprojected to a PCS of NAD 1927 UTM Zone 12N and were clipped to the extent of the TICA GPS dataset. They serve as mid-resolution benchmarks for evaluating the effects of scale and source variation on derived terrain metrics.

30-Meter and 90-Meter SRTM DEMs

The 30-meter and 90-meter Shuttle Radar Topography Mission (SRTM) DEMs were produced from radar interferometry data collected during a NASA mission in February 2000 (U.S. Geological Survey, 2009a, 2009b). The 30-meter version (1 arc-second) offers improved detail in U.S. territories, while the 90-meter version (3 arc-seconds) was developed for broader international use. Both datasets exhibit radar-specific artifacts such as voids and vegetation bias but offer valuable insights into how remotely sensed elevation products compare to more conventional DEM sources (Holmes et al., 2000). The srtm30m and srtm90m DEMs were reprojected to a PCS of NAD 1927 UTM Zone 12N and were clipped to the extent of the TICA GPS dataset. They serve as mid to low resolution benchmarks for evaluating the effects of scale and source variation on derived terrain metrics.

TICA GPS Dataset

The TICA ground validation dataset consists of 75 high-accuracy GPS elevation measurements collected within Timpanogos Cave National Monument. The points were collected by Dr. Michael Harman of the Department of Geology and Geography at West Virginia University, following a methodological framework inspired by Holmes et al. (2000) (Timpanogos Cave National Monument GPS Dataset, n.d.). The GPS point layer has a native PCS of NAD 1927 UTM Zone 12N. The GPS points serve as a benchmark for evaluating the vertical accuracy of each DEM used in the analysis. Since the points are high precision and have known spatial locations, they allow for direct comparisons between real-world elevations and DEM-derived values.

Spatial Analyses

All spatial analyses for this study were conducted using ArcGIS Pro 3.4.2 (ArcGIS, 2024). The initial step involved generating the custom 1-meter resolution dem1m raster using the "Topo to Raster" tool. The input for this tool was a digitized 2-foot polyline contour dataset covering the TICA area. The Topo to Raster tool uses the ANUDEM algorithm to generate hydrologically correct DEMs by interpolating elevation data while enforcing surface continuity and minimizing artificial depressions or flat areas. Although the tool was executed with default settings in this exercise, it offers additional options, such as integrating stream data, adjusting feature weights, and applying smoothing or tolerance parameters. These options can enhance flow realism and surface quality, particularly for hydrologic modeling and watershed applications (Esri, n.d.-a). The resulting dem1m raster served as one of six (6) DEMs examined in the analysis. The five remaining DEMs included lidar2m, ned10m, ned30m, srtm30m, and

srtm90m. Each DEM was standardized by reprojecting to a common PCS and clipping to the study area extent defined by the TICA GPS dataset.

From each DEM, slope and total curvature surfaces were derived utilizing tools from the Spatial Analyst toolbox in ArcGIS Pro. Slope was calculated in degrees, providing insight into terrain steepness and elevational change. Curvature was derived using the “Curvature” tool, which measures the overall convexity or concavity of the terrain in all directions and is useful for general terrain characterization. Although only total curvature was used in this analysis, the tool also offers profile and plan curvature outputs. Profile curvature is the curvature of the surface in the direction of slope and is important for modeling flow acceleration and erosion. While plan curvature is perpendicular to the slope direction and helps describe the convergence or divergence of surface flow, influencing soil moisture and hydrological patterns (Esri, n.d.-b). The complete derivative set includes elevation, slope, and total curvature for all six (6) DEMs. Resulting in eighteen (18) raster datasets that were used in subsequent analyses of local variability and error behavior.

In order to assess operational scale and local topographic complexity, a 3x3 neighborhood focal standard deviation was calculated for each of the eighteen (18) raster layers using the “Focal Statistics” tool in ArcGIS Pro. These focal rasters measured local variation by capturing the pixel-level standard deviation within a moving window. The mean local variation value for each focal raster was retrieved from the Symbology pane. These values provided insight into how much local variability is preserved or generalized at different spatial resolutions and with different source methods.

The “Extract Multi Values to Points” tool was used to sample elevation values from each of the six (6) DEMs at the locations of 75 high-accuracy GPS survey points collected within

TICA to evaluate dataset accuracy. For each DEM, absolute error values were calculated by subtracting the GPS-measured elevation from the DEM-derived elevation. These values were summarized in Excel to compute average, median, maximum, and minimum absolute elevation errors per dataset. Offering a standardized measure of vertical accuracy across models.

Finally, the relationship between slope and absolute elevation error was examined by plotting the extracted slope values against corresponding absolute elevation errors for each DEM. Scatterplots were created in Excel, and linear trendlines were fitted to each to derive R^2 values. These coefficients quantified the strength of association between terrain steepness and vertical error magnitude. Together, these spatial analyses provided a comprehensive foundation for evaluating the positive and negative aspects of each DEM relative to terrain modeling goals.

Results

DEM Derivation and Surface Representation

The six (6) DEMs and their derived slope and curvature surfaces reveal noticeable qualitative differences in terrain representation. Particularly in their alignment with known topographic features such as contour lines. High-resolution datasets such as the lidar2m and dem1m datasets offer the most detailed surface representations. Specifically, these datasets preserve subtle terrain undulations and finer landscape features. In comparison, the coarser srtm30m and srtm90m datasets exhibit generalized elevation forms and coarse pixelation, eliminating fine-scale terrain details. Also worth mentioning, the ned10m product displays a pronounced “waffling” artifact in its curvature output. Which is most likely a result of resampling the original 30-meter dataset to a finer resolution. These visual distinctions

emphasize that resolution alone does not determine DEM quality; the source data collection method and resampling procedures also play a key role.

Local Terrain Variability and Operational Scale

Quantitative assessment of spatial scale was performed using the "Focal Statistics" tool on each DEM and its terrain derivatives. Table 1 and Figure 1 summarize the mean local variation for elevation, slope, and curvature across all datasets. For elevation, srtm90m displayed the highest average local variation of 47.68373. Suggesting it captures broader topographic variability over a coarse scale. Slope values remained relatively stable across all resolutions, but trended highest in coarse datasets like srtm90m (6.68709) and srtm30m (6.50108). In contrast, the curvature variation was highest in the lidar2m dataset at 31.33053, indicating that curvature is a highly sensitive metric that quickly loses detail at coarser resolutions. These results reflect theoretical expectations. Elevation is best used for large-scale relief characterization, slope reflects mid-scale terrain shifts, and curvature captures localized surface complexity. Therefore, each DEM serves a unique analytical purpose, and no single dataset is universally superior.

Vertical Accuracy Compared to TICA GPS Dataset

In order to quantify vertical accuracy, elevation values were extracted from each parent DEM at 75 high-accuracy GPS point locations using the "Extract Multi Values to Points" tool. Table 2 displays the summary statistics for absolute elevation error. The dem1m dataset achieved the lowest average elevation error at 2.94 m, followed closely by lidar2m at 1.86 m, highlighting the advantage of high-resolution elevation products. Coarser datasets such as ned10m (31.83 m), ned30m (28.90 m), and srtm90m (27.13 m), exhibited significantly higher average errors. These

findings highlight the limitations of interpolated or radar-derived datasets in steep, complex environments such as the TICA study area. While coarser DEMs may suffice for general elevation trends, they fall short in applications requiring high-precision vertical positioning.

Slope-Error Relationships Across DEMs

Table 3 and Figures 2 through 4 explore the relationship between slope and elevation error by examining R^2 values for each DEM. These values reflect the degree to which slope influences vertical accuracy. The dem1m ($R^2 = 0.2045$) and ned30m ($R^2 = 0.1818$) datasets exhibited the strongest relationships, indicating that terrain steepness amplifies error magnitude. In contrast, the srtm90m ($R^2 = 0.0391$) and lidar2m ($R^2 = 0.0497$) datasets exhibited weaker correlations, suggesting more random or generalized error distributions. While a strong slope–error relationship might allow for error correction modeling. It also implies biased results in slope-sensitive applications like erosion modeling or flow direction estimation. This reinforces the need to consider terrain complexity and slope sensitivity when selecting an appropriate DEM for a given analysis.

Resolution Tradeoffs and Analytical Suitability

The collective results emphasize that no single DEM offers universal advantages across all spatial analyses. High-resolution datasets like dem1m and lidar2m excel in precision and detail, capturing fine-scale curvature and slope dynamics with minimal error. However, these datasets often require more storage space and processing power. Coarser DEMs such as srtm90m and ned30m are less detailed, but still useful for regional elevation trends or large-scale modeling efforts. Intermediate-resolution products like **ned10m** may seem to offer a balance, but can

suffer from resampling artifacts that degrade their reliability. Ultimately, the best DEM depends on the specific goals, acceptable error thresholds, and the scale of analysis. For fine-scale topographic or hydrologic applications, high-resolution datasets are preferred. For broader assessments, coarser products may be sufficient, provided their limitations are acknowledged and addressed accordingly.

Discussion

Each DEM evaluated in this study presents a unique balance of strengths and limitations, highlighting the trade-offs between spatial resolution, accuracy, and analytical utility. High-resolution surfaces such as lidar2m and dem1m were especially effective in preserving fine-scale terrain features and yielded the lowest elevation errors relative to the TICA GPS dataset. However, dem1m required manual digitization from 2-foot contours, and lidar2m, though automated, demands significant storage and computational resources. Coarser products like srtm30m and srtm90m proved more appropriate for capturing broad elevation patterns, but lacked the detail necessary for fine-scale analysis and exhibited greater vertical error. Meanwhile, ned10m, despite its intermediate resolution, showed a distinct “waffling” artifact from resampling 30-meter inputs, compromising its reliability for precision terrain modeling.

Results from the local variation analysis further reinforced the scale-dependent nature of each DEM. Elevation variability was highest in srtm90m, indicating its strength in capturing broad terrain relief. Curvature was highly sensitive to resolution, with lidar2m producing the highest variability and coarser DEMs rapidly losing detail. Slope values remained relatively consistent across resolutions, but peaked in the coarser datasets. These findings emphasize that

no single dataset performs best across all metrics. Each dataset serves a specific role depending on the desired scale of analysis and terrain complexity.

Vertical accuracy analysis revealed that elevation error often increases with slope, particularly in dem1m and ned30m. This slope–error relationship indicates that steeper terrain can amplify measurement biases in higher-resolution or resampled datasets. While such biases may be correctable through statistical calibration, they also signal caution for using these surfaces in slope-sensitive applications like hydrologic or erosion modeling. Ultimately, dataset selection must consider both resolution and source method. High-resolution DEMs are essential for localized, detail-rich analyses. Whereas coarser DEMs may suffice for regional modeling, their limitations must be acknowledged and managed appropriately.

References

ArcGIS, Version 3.4.2. Redlands, CA: EnvironmentalSystems Research Institute, Inc

Cao, C., & Lam, N. S.-N. (1997). Understanding the scale and resolution effects in remote sensing and GIS. In D. A. Quattrochi & M. F. Goodchild (Eds.), Scale in remote sensing and GIS (pp. 57–72). CRC Press.

Esri. (n.d.-a). Topo to Raster (Spatial Analyst)—ArcGIS Pro. Retrieved July 10, 2025, from https://pro.arcgis.com/en/pro-app/3.4/tool-reference/spatial-analyst/topo-to-raster.htm

Esri. (n.d.-b). Curvature (Spatial Analyst). ArcGIS Pro 3.4. Retrieved July 10, 2025, from https://pro.arcgis.com/en/pro-app/3.4/tool-reference/spatial-analyst/curvature.htm

Holmes, K. W., Chadwick, O. A., & Kyriakidis, P. C. (2000). Error in a USGS 30-meter digital elevation model and its impact on terrain modeling. Journal of Hydrology, 233(1–4), 154–173. https://doi.org/10.1016/S0022-1694(00)00229-8

McNeil, B. E., Jasper, J. D., Luchsinger, D. A., & Rainsmier, M. V. (2002). Implementation and application of GIS at Timpanogos Cave National Monument, Utah. Journal of Cave and Karst Studies, 64(1), 34–37.

Timpanogos Cave National Monument GPS Dataset (TICA). (n.d.). High-accuracy ground validation points collected by instructor for educational use.

U.S. Geological Survey (USGS). (2009a). Shuttle Radar Topography Mission (SRTM), 1 arc-second (~30m) resolution. https://www.usgs.gov/centers/eros/science/national-elevation-dataset

U.S. Geological Survey (USGS). (2009b). Shuttle Radar Topography Mission (SRTM), 3 arc-second (~90m) resolution. https://www.usgs.gov/centers/eros/science/national-elevation-dataset

U.S. Geological Survey (USGS). (2009c). National Elevation Dataset (NED), 1 arc-second (~30m) resolution for the United States. https://www.usgs.gov/core-science-systems/ngp/3dep

U.S. Geological Survey (USGS). (2009d). National Elevation Dataset (NED), 1/3 arc-second (~10m) resolution for the United States. https://www.usgs.gov/core-science-systems/ngp/3dep

Utah Automated Geographic Reference Center (AGRC). (2008). AGRC 2 meter LiDAR dataset. Horizons, Inc. Acquired October 12, 2006 – May 15, 2007. https://agrc.utah.gov

Wilson, J. P. (2012). Digital terrain modeling. Geomorphology, 137(1), 107–121. https://doi.org/10.1016/j.geomorph.2011.06.027

Woodcock, C. E., & Strahler, A. H. (1987). The factor of scale in remote sensing. Remote Sensing of Environment, 21(3), 311–332. https://doi.org/10.1016/0034-4257(87)90015-0

Table 1. Mean local variation values derived from focal statistics for each DEM and its derivatives.

Mean Local Variation Across DEMs Derived from Focal Raster Analysis			
Parent DEM	**DEM Derivative Name**	**Focal Raster Name**	**Mean Local Variation**
dem1m	Parent DEM	dem1m_focal	0.81298
	dem1m1_slp	dem1m1_slp_focal	3.19551
	dem1m_totalcurv	dem1m_totalcurv_focal	20.04761
srtm90m	Parent DEM	srtm90m_focal	47.68373
	srtm90m_slp	srtm90m_slp_focal	6.68709
	srtm90m_totalcurv	srtm90m_totalcurv_focal	0.82648
srtm30m	Parent DEM	srtm30m_focal	17.70421
	srtm30m_slp	srtm30m_slp_focal	6.50108
	srtm30m_totalcurv	srtm30m_totalcurv_focal	2.54713
ned30m	Parent DEM	ned30m_focal	19.36088
	ned30m_slp	ned30m_slp_focal	6.46242
	ned30m_totalcurv	ned30m_totalcurv_focal	2.9252
ned10m	Parent DEM	ned10m_focal	6.92218
	ned10m_slp	ned10m_slp_focal	5.28067
	ned10m_totalcurv	ned10m_totalcurv_focal	6.11388
lidar2m	Parent DEM	lidar2m_focal	1.67403
	lidar2m_slp	lidar2m_slp_focal	3.67702
	lidar2m_totalcurv	lidar2m_totalcurv_focal	31.33053

The Mean Local Variation was obtained using the “Focal Statistics” tool in ArcGIS Pro. The mean value was retrieved from the Symbology pane. For clarity, all values have been rounded to five (5) decimal places.

Table 2. Summary of error statistics derived from GPS-to-DEM elevation comparisons across all six elevation datasets.

Elevation Summary Error Statistics				
Parent DEM	**Average Elevation Error**	**Median Elevation Error**	**Maximum Elevation Error**	**Minimum Elevation Error**
dem1m	2.938684612	2.0740439	17.9618926	0.0729902
srtm90m	27.12666667	21.117	96.154	0.594
srtm30m	17.67616	13.708	72.154	0.594
ned30m	28.90080625	22.1580537	110.3669512	0.1290488
ned10m	31.82920351	25.2180439	120.8040293	0.2189561
lidar2m	1.86344	1.259	8.519	0.074

Error statistics are absolute and were computed by comparing DEM elevation values to GPS survey heights from the Timpanogos Cave National Monument (TICA) dataset.

Table 3. Summary of error relationships between slope and absolute elevation error across all six elevation datasets.

Slope vs. Absolute Elevation Error - R^2 Values			
Parent DEM	**Resolution (m)**	**Source**	**R^2**
dem1m	1	Topo to Raster from vector elevation data	0.2045
srtm90m	90	SRTM	0.0391
srtm30m	30	SRTM	0.1016
ned30m	30	USGS NED	0.1818
ned10m	10	USGS NED	0.1686
lidar2m	2	LiDAR	0.0497

R^2 values were generated from a linear trendline and represent the strength of the relationship between slope and absolute elevation error. Errors were calculated by comparing DEM-derived elevations to GPS survey heights from the Timpanogos

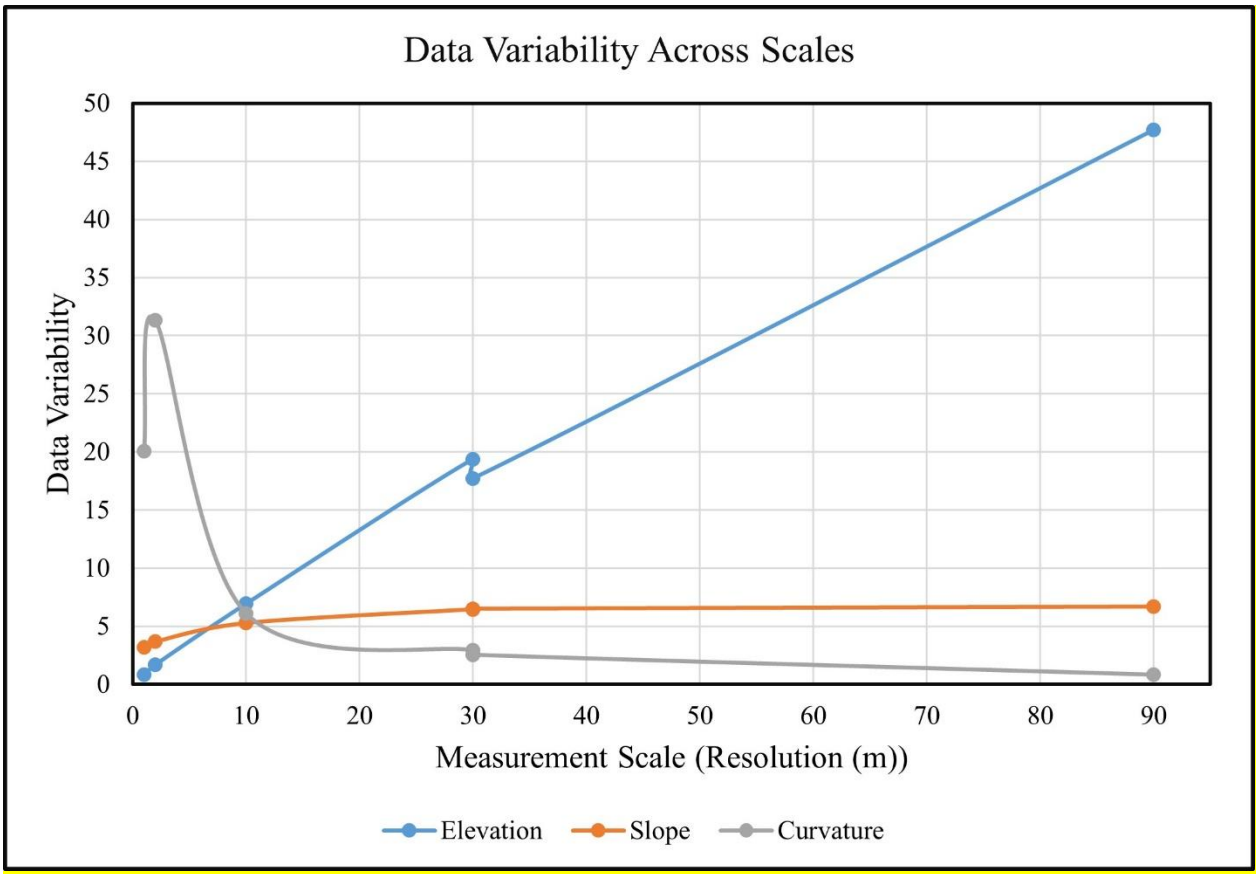

Figure 1. Graph showing the average local variability in elevation, slope, and curvature across DEMs of different spatial resolutions.

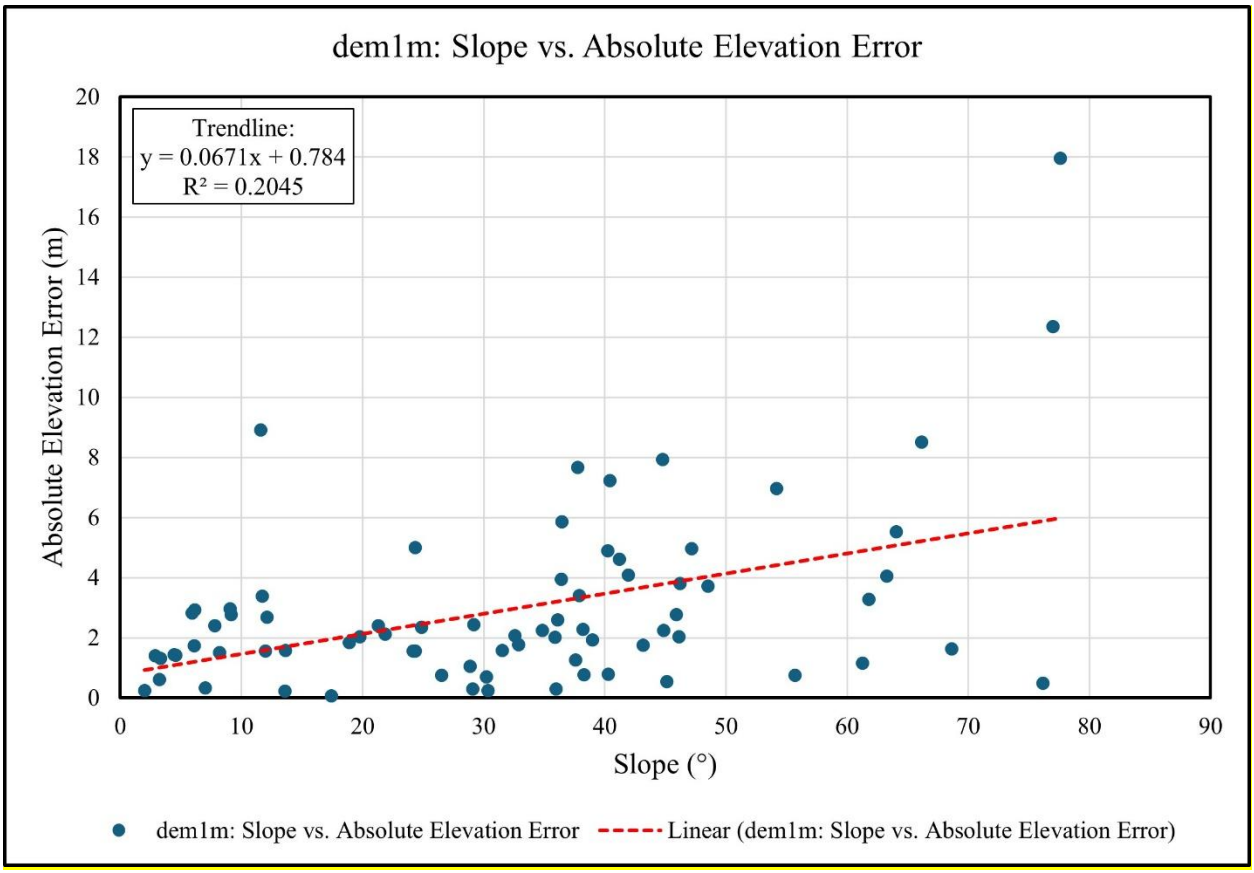

Figure 2. Scatterplot showing the relationship between slope and absolute elevation error for the 1-meter resolution DEM constructed using the Topo to Raster tool.

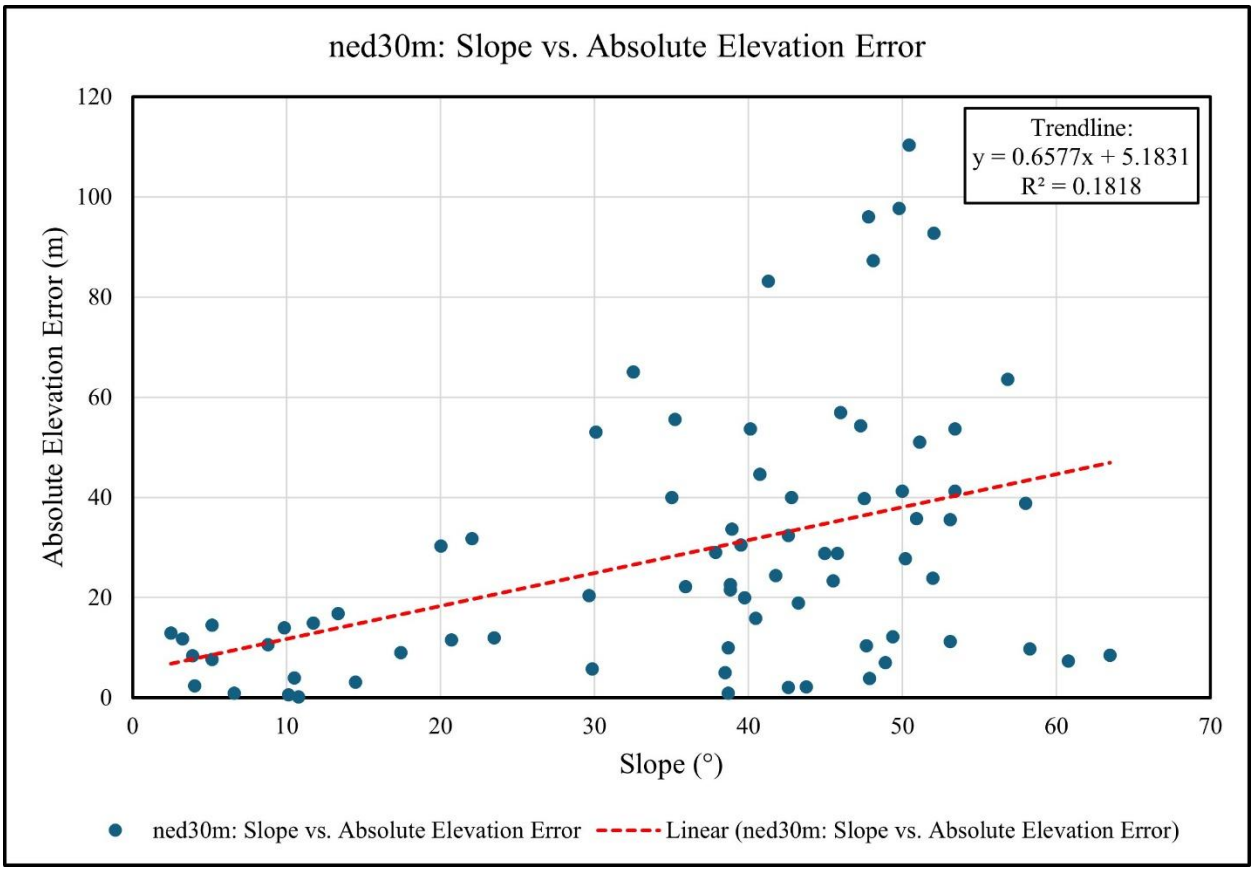

Figure 3. Scatterplot showing the relationship between slope and absolute elevation error for the USGS 30-meter NED DEM.

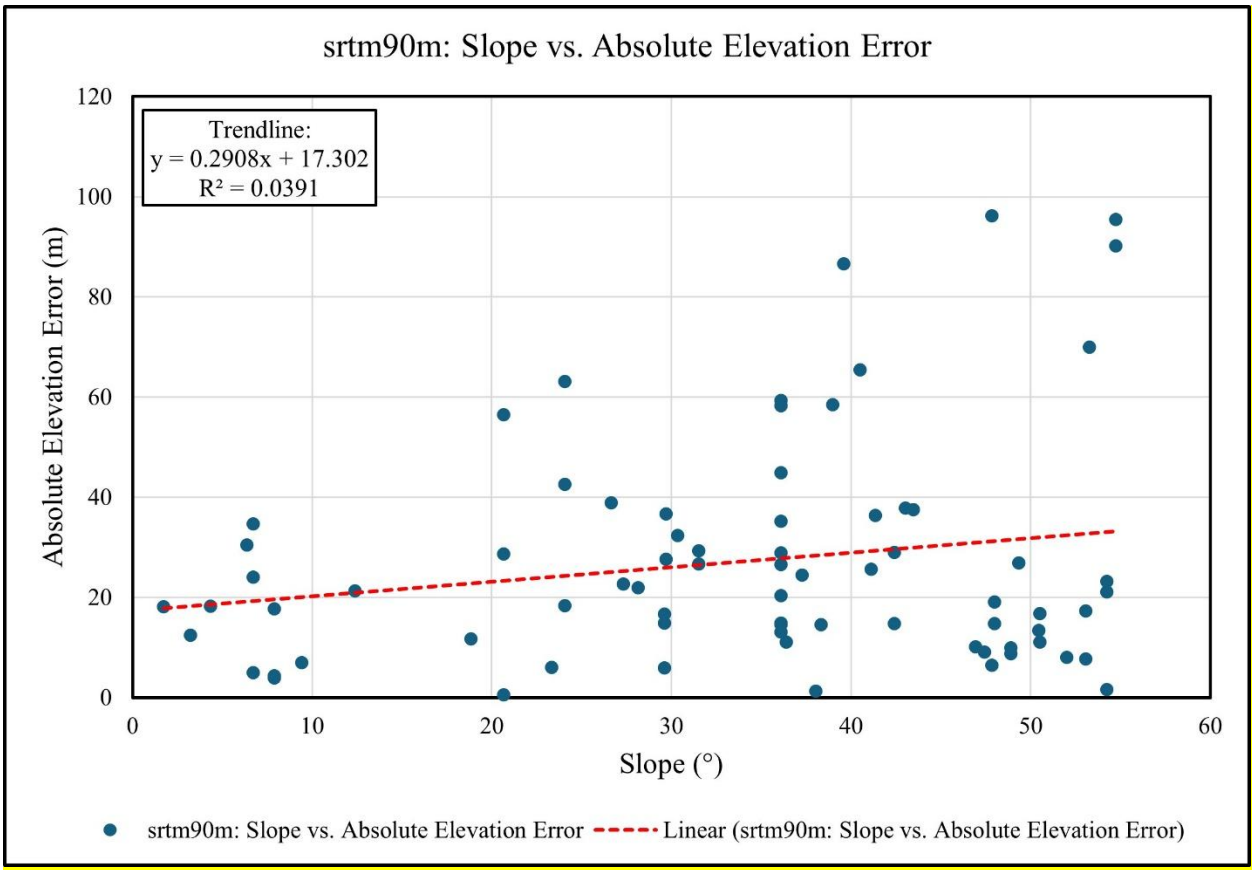

Figure 4. Scatterplot showing the relationship between slope and absolute elevation error for the 90-meter SRTM DEM.